

Section 1:

10 Structured BI Prompts & Evaluation

1. Data Cleaning Steps

- **Prompt:** Act as a data analyst. I have an operational dataset with columns Date, Processing_Center, and Waste_Mass_KG. Write a Python pandas script to find missing rows, drop duplicate entries, and convert the Date column into a standard datetime format.
- **AI Tool Used:** Gemini
- **Output Summary:** Provided a clean, 4-line pandas script using `.drop_duplicates()`, `.isna()`, and `pd.to_datetime()`.
- **Evaluation:** Highly useful. The code correctly targets the structural issues of raw operational data without formatting bugs.

2. Defining 3 KPIs

- **Prompt:** You are a senior supply chain consultant. For an agricultural processing firm tracking raw crop waste, define 3 core operational KPIs that connect physical waste mass to financial value losses.
- **AI Tool Used:** Gemini
- **Output Summary:** Recommended Aggregate Financial Loss (INR), Waste Mass Ratio (%), and Net Biomass Yield per Center.
- **Evaluation:** Excellent alignment. These metrics match the exact needs of our HimShakti dashboard framework.

3. Writing an Executive Summary

- **Prompt:** Write a 100-word executive summary for a COO based on this finding: Total operational loss is 17,203,520 INR, and Herbal Extract contributes to 56.4% of it. Focus heavily on financial risk.
- **AI Tool Used:** ChatGPT
- **Output Summary:** Generated a direct paragraph outlining the 17.2M INR deficit and warning about margin drops from the extraction line.
- **Evaluation:** Highly effective. It focused on the primary business problem without adding filler text.

4. Generating SQL Queries

- **Prompt:** Given a database table named center_logs with columns center_id, material_type, and loss_inr, write a SQL query to calculate the total loss for 'Herbal Extract' grouped by center.
- **AI Tool Used:** Gemini
- **Output Summary:** Provided a standard SELECT center_id, SUM(loss_inr) FROM center_logs WHERE material_type = 'Herbal Extract' GROUP BY center_id; query.
- **Evaluation:** Completely accurate. The syntax executes perfectly for rapid database segmentation.

5. Recommending a Chart Type

- **Prompt:** I need to display the breakdown of financial losses across three distinct crop categories (Ginger, Turmeric, Herbal Extract). Recommend the best chart type for a business audience and explain why.
- **AI Tool Used:** Claude
- **Output Summary:** Recommended a Donut Chart or a Horizontal Bar Chart to easily show share-of-whole composition.
- **Evaluation:** Accurate. This confirms that the donut chart used in our Week 4 dashboard layout was the right choice.

6. Interpreting a Correlation

- **Prompt:** Explain what it means if our data shows a high positive correlation between Inbound_Weight_KG and Waste_Mass_KG at Center-A, but a low correlation at Center-B. Keep it simple.
- **AI Tool Used:** Gemini
- **Output Summary:** Explained that Center-A's waste grows linearly with volume, while Center-B has more stable, efficient processes.
- **Evaluation:** Incredibly insightful. This distinction directly explains the variance we observed in our physical logs.

7. Detecting Data Quality Issues

- **Prompt:** Review this data trend: Center-A records a flat 0% waste mass for seven consecutive days during high-volume inbound periods. What data quality issues should I investigate?
- **AI Tool Used:** ChatGPT
- **Output Summary:** Flagged likely hardware issues like stuck weighbridge scales or broken data entry logging pipelines.
- **Evaluation:** Practical. It gave us highly realistic check-points for our factory-floor logistics audit.

8. Writing a Slide Headline

- **Prompt:** Turn this data point into a short, punchy, action-oriented slide headline: 'Herbal Extract extraction losses account for 17.2 Million INR in total waste.' Limit it to 10 words.
- **AI Tool Used:** Claude
- **Output Summary:** "Herbal Extract Extraction Drives 56.4% of Total Financial Loss"
- **Evaluation:** Perfect. It fits the presentation text constraints while calling out the main driver clearly.

9. Drafting a Business Recommendation

- **Prompt:** Based on the finding that Center-A handles high volumes but generates major processing waste mass, write a 2-step strategic recommendation for the plant director.
- **AI Tool Used:** Gemini
- **Output Summary:** Suggested performing immediate machinery recalibrations and running training based on Center-B's workflows.
- **Evaluation:** Actionable and realistic. It matches our executive mitigation strategy perfectly.

10. Explaining a Technical Concept

- **Prompt:** Explain what an API data pipeline automated storage loop is to a traditional non-technical operations manager who only uses physical paper logs.
- **AI Tool Used:** ChatGPT
- **Output Summary:** Explained it as a digital conveyor belt that moves scale metrics straight to reports without manual entry.
- **Evaluation:** Great translation. It uses a helpful analogy to explain the concept without technical jargon.

Section 2:

3-Variation Prompt Comparison

- **Task Tested:** Writing a Board-level Executive Summary
- **Variation 1 — Minimal:** "Write an executive summary of our Q2 dashboard data."
 - **Output:** Generates a highly generic template filled with placeholder brackets since no context or metrics were provided.
- **Variation 2 — Context-Rich:** "You are a BI analyst. The dashboard shows a total loss of 17,203,520 INR with Herbal Extract causing 56.4%. Write a 100-word executive summary for our CEO."
 - **Output:** Incorporates the accurate metrics, but lacks logical structure and runs as a dense wall of text.
- **Variation 3 — Structured + Role:** "Act as a senior BI analyst. Write a 100-word executive summary for the COO regarding Q2 yield leakages. Structure it in exactly two bullet points: 1. Main Finding, 2. Financial Impact. Use direct, business-friendly language and avoid technical jargon."
 - **Output:** Delivers a punchy, perfectly formatted, executive-ready breakdown completely tailored to the data points.

Comparative Evaluation

Variation 3 worked best because it combined a clear professional role with strict output constraints. By restricting the format to distinct bullet points and outlawing technical jargon, it forced the model to prioritize critical business impacts over generic filler sentences.

The prompt pattern that made the biggest difference was **Constraint-Based Role Prompting**. Giving the AI an explicit target audience and strict structural limitations is what separates generic text generation from high-value, factual business intelligence insights.